

NAME: Sudharsan R

Rollno: 240701543

**Experiment 1: COMPARATIVE STUDY OF LOW-CODE
DEVELOPMENT VERSUS TRADITIONAL
APPLICATION DEVELOPMENT APPROACHES**

AIM:

To study and compare the **traditional application development approach** with the **low-code development approach** based on development process, coding effort, development time, cost, maintenance, and overall efficiency using **Zoho Creator** as an example of a low-code platform.

INTRODUCTION TO TRADITIONAL APPLICATION DEVELOPMENT:

Traditional application development is the process of creating software by writing source code using programming languages such as Java, Python, C#, or PHP. Developers manually design the user interface, write business logic, connect databases, test the application, and deploy it.

This approach requires strong programming knowledge and usually involves a team of software developers, testers, database administrators, and project managers. Although it offers maximum flexibility and customization, it takes more time and effort to build and maintain applications.

Features

- Requires extensive coding
- High flexibility and customization
- Longer development time
- Skilled programmers are required
- Higher development and maintenance cost
- Suitable for complex enterprise applications

INTRODUCTION TO LOW CODE DEVELOPMENT:

Low-code development is a modern software development approach that allows applications to be built with minimal hand-written code. It uses visual interfaces, drag-and-drop components, pre-built templates, and automated workflows.

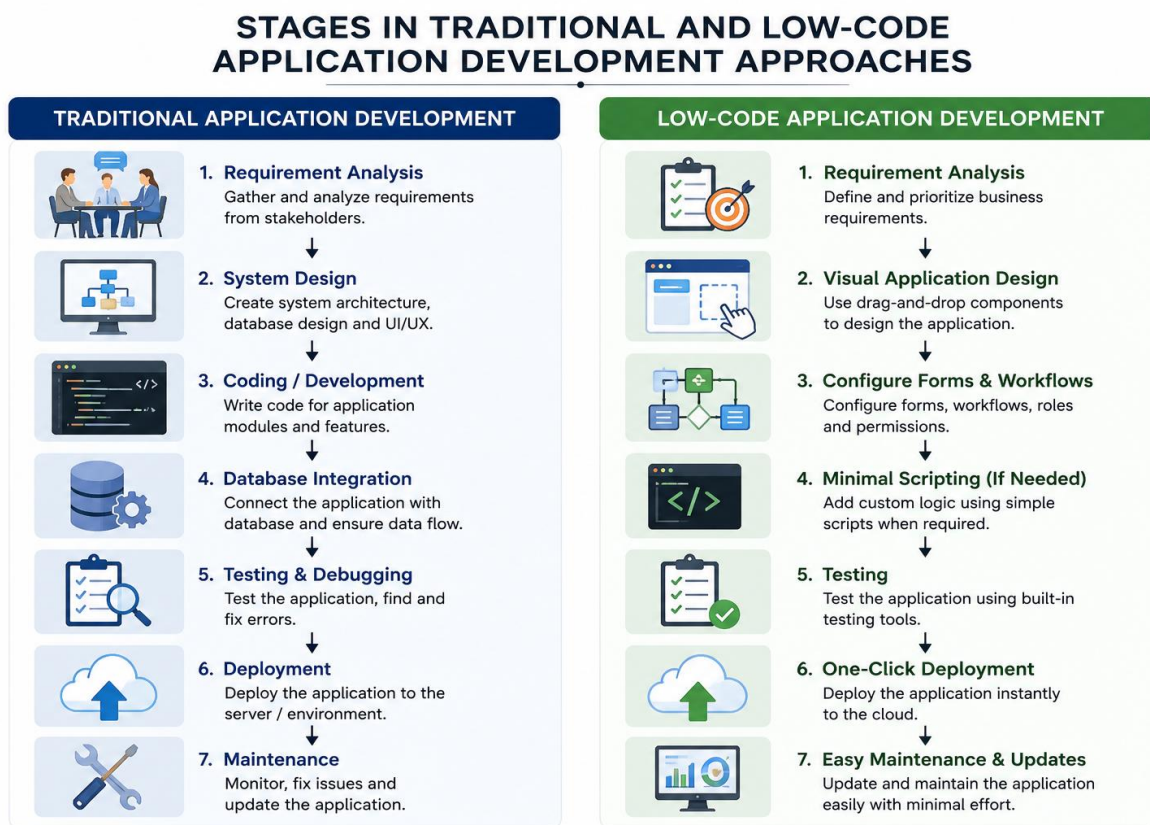
Zoho Creator is a popular low-code platform that enables users to develop web and mobile applications quickly. It provides visual tools for designing forms, workflows, reports, and dashboards. Small amounts of scripting (Deluge language) can be added when advanced functionality is needed.

Low-code platforms reduce development time, simplify maintenance, and enable even non-programmers (citizen developers) to create business applications.

Features:

- Minimal coding required
- Drag-and-drop interface
- Faster application development
- Lower development cost
- Easy deployment and maintenance
- Suitable for business applications and rapid development

DIAGRAM FOR STAGES IN TRADITIONAL AND LOW-CODE DEVELOPMENT:



COMPARISON BETWEEN TRADITIONAL AND LOW-CODE DEVELOPMENT:

Parameter	Traditional Development	Low-Code Development (Zoho Creator)
Coding	Extensive manual coding	Minimal coding
Development Time	Weeks or months	Hours or days
Cost	High	Low
Speed	Slow	Fast
Skills Required	Professional programmers	Basic programming knowledge
Customization	Very High	Moderate
Maintenance	Difficult	Easy
Deployment	Complex	Simple
Best Used For	Large and complex applications	Business and internal applications

CONCLUSION:

Traditional application development provides maximum flexibility and control but requires more time, cost, and programming expertise. Low-code development platforms like **Zoho Creator** simplify application development by using visual tools and reusable components, enabling faster development, easier maintenance, and lower costs. Low-code is ideal for rapidly building business applications, while traditional development is preferred for highly customized and complex software systems.